Excellia RAG Project

Progress Report

Model Testing & Performance Optimization

July 30, 2026

Executive Summary

This report summarizes the progress made on the Excellia RAG system, including model testing, performance optimization, and architectural decisions.

1. Model Testing & Results

Model	Parameters	Response Time	Status	Verdict
Qwen-2.5:7B-Instruct	7B	>200 seconds	■ Too Heavy	Rejected
Phi-3 Mini	3.8B	~100 seconds	■■ Acceptable	Borderline
Llama 3.2:3B	3B	~20 seconds	■ Good	Selected

Key Optimizations

• Reduced DENSE_TOP_K from 25 to 15 (40% reduction)

• Reduced SPARSE_TOP_K from 25 to 15 (40% reduction)

• Reduced FUSED_TOP_N from 15 to 8 (47% reduction)

• Disabled reranking for speed (100% reduction)

• Optimized confidence threshold from 0.1 to 0.005

2. System Architecture

- 1. User Question Input
- 2. Embedding Generation (FastEmbed - BAAI/bge-small-en-v1.5)
- 3. Dense Search (Qdrant - Top 15)
- 4. Sparse Search (BM25 - Top 15)
- 5. Hybrid Fusion (RRF - K=60, Top 8)
- 6. Reranking (DISABLED for performance)
- 7. Context Building (Top 8 chunks)
- 8. LLM Generation (Ollama - llama3.2:3b)
- 9. Answer Output

Architecture Flow

User Question → Embedding → Dense Search → Sparse Search → RRF Fusion → Context → LLM → Answer

3. Performance Metrics

Metric	Value	Status
Embedding Model	BAAI/bge-small-en-v1.5	CPU Optimized
Vector Database	Qdrant	Operational
LLM Model	llama3.2:3b	Selected
Chunk Size	300 tokens	Optimal
Average Response Time	~20 seconds	Good
Chunks Indexed	10,869	Successful

Model Comparison

• Qwen-2.5:7B: 200s response time, too heavy for CPU

• Phi-3 Mini: 100s response time, borderline performance

• Llama 3.2:3B: 20s response time, optimal choice

4. Key Learnings

• • Larger models (7B) are not suitable for CPU-only deployment

• • 3B-4B parameter models provide optimal balance

• • Response time improved from 200s to 20s (10x faster)

• • Disabling reranking significantly improves performance

• • Confidence threshold needs careful tuning per document type

• • FastEmbed provides efficient CPU embeddings without PyTorch

• • Chunk size and overlap parameters affect retrieval quality

• • RRF fusion effectively combines dense and sparse search results

5. Challenges & Solutions

Challenge	Issue	Solution
Model Size	7B models too large for CPU	Switched to 3B models (llama3.2:3b)
Response Time	Initial response time >200s	Reduced parameters and disabled reranking → 20s
Confidence Threshold	UPI documents found but not reliable	Lowered threshold from 0.1 to 0.005
Embedding Speed	Slow embedding generation	Used FastEmbed with progress tracking
Memory Usage	High memory consumption	Optimized chunk sizes and top_k values

6. Test Results Summary

Query Type	Result	Status
Excellia Trade	■ Found relevant information	High Confidence
UPI (after threshold fix)	■ Found relevant information	Medium Confidence
Documentary Credits	■ Found relevant information	High Confidence
SWIFT Processing	■ Found relevant information	High Confidence

7. Conclusion

The Excellia RAG system has successfully evolved through multiple iterations of model testing and optimization.

Key Achievements:

• Working RAG pipeline with hybrid search

• Successful document ingestion (Excellia Trade + UPI documents)

• FastEmbed for CPU-optimized embeddings

• Response time reduced by 10x (200s → 20s)

• Confidence threshold tuned for better retrieval

The system is now ready for testing with real users and can be further optimized based on usage patterns.

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